

High Level Summary

THE PROPOSITION

- A smartphone camera measures blood pressure and pulse rate. Micro colour changes in facial skin (remote photoplethysmography) analysed by proprietary AI.
- CE Class IIa EU MDR certified for BP and pulse. Currently in pilot with NHS
- FDA clearance in progress.

WHERE THE VALUE SITS

- The science is proven with six patents secured.
- A data model of 25,000+ paired clinical recordings from 12,500 patients.
- Both systolic and diastolic to clinical grade.
- The value needs to be unlocked, getting this into the hands of 100 million people, not 100 thousand.
- The current solution doesn't scale, this is a platform problem.

THE TECHNICAL CHALLENGE

Parallel improvements required:

- Scale without breaking what's in production today
- Every change traceable for regulators
- Deliver with a small focused team

Top Five Areas

	T O D A Y	T A R G E T	A G E N T I C A I
Reduce cost	10p+ per assessment Cold container per request Economics break at scale	<1p per assessment Warm inference pool (AKS) On-device signal extraction	Cost thresholds monitored via Grafana alerting Ops agent correlates spikes with known change
Improve performance	No latency guarantee, users wait for model All compute in cloud Video transmitted from device	Async queue, warm pool Horizontal autoscaling Signal extracted on-device Metal/Vulkan GPU pipeline	Grafana alerting detects performance degradation ML Validation agent monitors output for model drift
Release safely at pace	Manual release gates No formal API contract Test evidence ad hoc	Automated pipeline Immutable artefacts Every sprint a release candidate	Compliance agent gates PRs Code Review agent on every PR Doc agent drafts changelogs
Observability	Limited visibility into production failures No mobile SNR telemetry No root cause analysis of failure	Grafana + Prometheus for metrics Loki for logs Sentry for error tracking Signal quality reported per assessment	Ops agent reads metrics Creates GitHub Issues from alerts AI surfaces, humans decide
Governance & planning	Decisions undocumented Sprint cadence informal Traceability manual	ADRs before code 1-week sprints, DoD with regulatory gates eQMS integrating with development process	Compliance agent enforces traceability at PR time Governance visible continuously